Unit 3, Calculus

Lecture 1: Approximation, Error, and the Idea of Change

Goal: Understand how calculus formalizes change and accumulation, and how it connects discrete models to continuous behavior in data science.
Motivation: Real-world systems are rarely solvable in closed form. We therefore construct simplified, approximate models that are easier to analyze and compute. Approximation underlies nearly all of scientific computing.

1. General Strategy

In mathematical and scientific computing, approximation plays a crucial role in solving complex problems. Complexity can arise from dimensionality, the nature of the equations, or variable interactions. One fundamental observation is that solving a similar but simpler problem can reveal insights into the solution of the more complicated one.
Common simplifications include:

- **Continuous → Discrete:** Differential equations are discretized using methods such as finite differences, which approximate derivatives with differences.
- **Discrete → Continuous:** Discrete data can be approximated by continuous models using interpolation or regression.
- **Differential → Algebraic:** Differential equations can be transformed into algebraic ones via discretization or integral transforms.
- **Nonlinear → Linear:** Nonlinear systems can be locally approximated by linear models near equilibrium points.

1.1 Example: Computing the Surface Area of the Earth

$$A = 4\pi r^2$$

- **Modeling Earth as a Sphere:** The Earth is approximated as a perfect sphere, ignoring its oblate shape and surface irregularities.
- **Empirical Measurements:** The radius r is based on imperfect empirical averages (e.g., $r \approx 6371$ km).
- **Truncating π:** Since π is irrational, we use a finite approximation such as $\pi \approx 3.14159$.
- **Finite Precision Arithmetic:** Computer arithmetic introduces rounding error.

These small assumptions make the problem computationally feasible, but also introduce error. If such approximations are reused, their errors propagate.

1.2 Sources of Error

Errors generally fall into three main categories:
1. Errors in the Inputs

- Measurement precision: instruments have limited resolution.

- Examples:
  - A ruler measures to the nearest millimeter.
  - A thermometer rounds to the nearest degree Celsius.

2. Errors in the Model

- Simplifications: modeling Earth as a sphere, approximating continuous equations by discrete analogs.
- Discretization introduces error depending on the step size.

3. Floating Point Errors

- Finite representation of real numbers leads to rounding errors.
- Example: Adding a very small number to a very large one may lose the small contribution.
- Repeated operations can accumulate error.

2. Mathematics of Error

To quantify error, we define:
Absolute Error:

$$E_a = |x_{\text{true}} - x_{\text{approx}}|$$

Relative Error:

$$E_r = \frac{|x_{\text{true}} - x_{\text{approx}}|}{|x_{\text{true}}|}$$

and often as a percentage:

$$E_{\%} = 100 \times E_r$$

2.1 Example: Approximating $\sin(\pi/8)$

Suppose we wish to compute $\sin(\pi/8)$ but choose an easier similar problem.
Simplifications:

- Replace $x = \pi/8$ with $\hat{x} = 3/8$.
- Replace $f(x) = \sin(x)$ with the small-angle approximation $\hat{f}(x) = x$.

True result:

$$f(x) = \sin(\pi/8) \approx 0.3827$$

Approximation:

$$\hat{f}(\hat{x}) = \hat{x} = 0.3750$$

Then:

$$E_a = |0.3827 - 0.3750| = 0.0077, \qquad E_r = \frac{0.0077}{0.3827} \approx 0.0201$$

So this approximation yields roughly a 2% relative error.

3. From Approximation to Change

The central question of calculus: *How does one quantity respond to a change in another?*

- Average rate of change:
$$v_{\text{avg}} = \frac{\Delta x}{\Delta t}$$
- As $\Delta t \to 0$, the secant line approaches the tangent line.
- The instantaneous rate of change defines the derivative.

Demonstration: Plot distance vs. time; draw secant lines for smaller Δt. As Δt shrinks, slope stabilizes to tangent slope.

4. The Concept of a Limit

As $\Delta t \to 0$, we want to formalize "approach" without necessarily reaching the point.

$$\lim_{x \to a} f(x) = L$$

means $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a.
Visualization: For $f(x) = x^2$, zooming near $x = 1$ shows near-linearity (local linearity).

Lecture 2: Limits, Continuity, and Differentiation Rules

2.1 Limits and Continuity

Definition:

$$\lim_{x \to a} f(x) = L$$

means as x gets close to a, $f(x)$ gets close to L.
Continuity at a point $x = a$:

1. $f(a)$ is defined.
2. $\lim_{x \to a} f(x)$ exists.
3. They are equal: $\lim_{x \to a} f(x) = f(a)$.

Examples:

- $f(x) = x^2$, continuous.
- $f(x) = \frac{x^2-1}{x-1}$, undefined at $x = 1$, but $\lim_{x \to 1} f(x) = 2$.
- Step functions, discontinuous (jump).

One-sided limits:

$$\lim_{x \to a^-} f(x), \qquad \lim_{x \to a^+} f(x)$$

If both exist and agree, the overall limit exists.
Limit at infinity:

$$\lim_{x \to \infty} f(x) = L, \quad \text{e.g. } \lim_{x \to \infty} \frac{1}{x} = 0.$$

2.2 The Derivative as a Limit

Definition:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Interpretations:

- Geometric: slope of the tangent line.
- Physical: instantaneous rate of change.
- Analytical: coefficient of best local linear approximation.

Example:

$$f(x) = x^2 \implies f'(x) = \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h} = 2x.$$

Multiple notations:

$$f'(x) = \frac{df}{dx} = \frac{d}{dx} f(x).$$

2.3 Differentiability and Continuity

- Differentiability implies continuity.
- Not all continuous functions are differentiable.
- Example: $f(x) = |x|$ continuous but not differentiable at $x = 0$.

2.4 Basic Derivative Rules

Constant and Power Rules:

$$(c)' = 0, \qquad (cf)' = cf', \qquad (x^n)' = nx^{n-1}.$$

Sum and Difference Rules:

$$(f+g)' = f' + g', \qquad (f-g)' = f' - g'.$$

Example: $f(x) = x^3 + 5x \Rightarrow f'(x) = 3x^2 + 5$.

2.5 Product and Quotient Rules

Product Rule:

$$(fg)' = f'g + fg'.$$

Quotient Rule:

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}, \qquad g(x) \neq 0.$$

Examples:

1. $f(x) = x^2 \sin x \Rightarrow f'(x) = 2x \sin x + x^2 \cos x$.
2. $f(x) = \dfrac{x^2+1}{x} \Rightarrow f'(x) = \dfrac{x^2-1}{x^2}$.

2.6 The Chain Rule

Used for compositions of functions.
Formula:

$$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x).$$

Interpretation: Multiply the outer derivative by the inner derivative. Each layer of computation contributes one factor.
Examples:

1. $y = \sin(x^2) \Rightarrow \dfrac{dy}{dx} = \cos(x^2) \cdot 2x.$

2. $y = e^{3x+1} \Rightarrow \dfrac{dy}{dx} = e^{3x+1} \cdot 3.$

3. $y = (5x^3 - 2)^4 \Rightarrow \dfrac{dy}{dx} = 4(5x^3 - 2)^3 \cdot 15x^2.$

Geometric meaning: The derivative of a composition represents how sensitivity compounds through nested relationships, a key principle behind backpropagation in machine learning.

Second Derivatives and Concavity

The second derivative measures how the rate of change itself is changing:

$$f''(x) = \lim_{h\to 0} \frac{f'(x+h) - f'(x)}{h}.$$

Examples:

- Constant function: $f''(x) = 0$.
- Linear function: $f''(x) = 0$.
- Quadratic function: $f''(x) = 2$.

Concavity:

- $f''(x) > 0$, concave up.
- $f''(x) < 0$, concave down.
- $f''(x) = 0$ with sign change, inflection point.

Example: $f(x) = x(x+2)(x-2)$ is concave down for $x < 0$, concave up for $x > 0$, with inflection at $x = 0$.

Numerical Derivatives

Computers cannot take exact limits, so derivatives must be approximated.
Finite difference methods:

- **Forward difference:** $f'(x) \approx \dfrac{f(x+h) - f(x)}{h}$
- **Backward difference:** $f'(x) \approx \dfrac{f(x) - f(x-h)}{h}$

- **Central difference:** $f'(x) \approx \dfrac{f(x+h) - f(x-h)}{2h}$

The central method is most accurate because it uses both sides of x.
Second derivative:

$$f''(x) \approx \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}.$$

Precision trade-offs:

- Large h: poor accuracy (approximation error).
- Small h: round-off or noise error dominates.

Noisy Derivatives and Rolling Averages

When data is noisy, derivative estimates can fluctuate wildly. A smoothing step can help.
Rolling average formula:

$$\text{Avg}(y_i) = \frac{1}{2N+1} \sum_{j=-N}^{N} y_{i+j}$$

This averages over nearby points to reduce noise.
Trade-off: A large window smooths noise but may blur important features. A small window preserves detail but leaves more noise. Selecting N depends on the application.